Jacob Kruse

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LinkedIn | GitHub | Personal Website

Employment

Senior Associate Robotics Engineer
L3Harris | 07/06/2026 - Present

Develop manipulation algorithms for the maritime robotics systems with deployment in Isaac Sim and conduct state-of-the-art robotics research

Graduate Research Assistant
THINC Lab | 08/14/2024 - 05/08/2026

Contributed to research in AI by working with robotic manipulators, applying computer vision models, implementing simulations, and integrating with ROS2

R&D Engineering Intern
IPG Photonics | 05/19/2025 - 07/25/2025

Developed automated R&D test stations, conducted tests, and recommended possible improvements to the laser products based on analysis of the data

Robotic Programming Intern
Lincoln Electric Automation | 05/15/2023 - 08/04/2023

Developed automation cells and programmed FANUC robots for diverse industrial tasks including welding, material handling, and vision inspection

Gas Distribution Engineering Intern
MidAmerican Energy | 05/23/2022 - 08/18/2022

Tested, troubleshot, and fixed natural gas pipelines to ensure compliance with cathodic protection standards

Civil Engineering Intern
IMEG | 05/2021 - 08/2021

Completed fieldwork on several civil projects and tested concrete, asphalt, and soil in the lab

Education

Master of Science in Computer Science
University of Georgia | 08/2024 - 06/2026

Researched reinforcement learning and its application to physical robotic systems

Bachelor of Science in Electrical Engineering
Western Illinois University | 08/2020 - 05/2024

Double Minor in Mathematics and Physics
Achieved a 3.91 GPA and received Summa Cum Laude and Electrical Engineering Scholar Awards

High School Diploma
Orion High School | 08/2016 - 05/2020

Achieved a 4.0 GPA and received Illinois State Scholar Award

Additional Experience

Master's Thesis Project
University of Georgia | 08/2025 - 05/2026

Developed a recycling sortation system that implemented reinforcement learning. Leveraged a KUKA collaborative robot for manipulation and YOLO instance segmentation for state estimation

Autonomous Vehicle Project
Western Illinois University | 08/2023 - 05/2024

Worked as the leader of the Navigation Team for my Senior Design Project. Led the design and integration of the vehicle guidance system utilizing LiDAR, Camera, GPS, and IMU sensors with ROS2

Freshman Basketball Coach
Orion High School | 10/2022 - 02/2024

Acted as the head coach of the Orion Freshman basketball team for 2 seasons

Skills

Programming	Software	Hardware	Miscellaneous
Python	KUKA OS	LiDAR	Wiring
C++	FANUC OS	RGBD Camera	Soldering
ROS2	SolidWorks	GPS/GNSS	Troubleshooting
Ubuntu/Linux	Fusion 360	IMU	Quick Learner
Nav2/MoveIt	NI Multisim	Arduino	Strong Leader
MySQL	Office 365	Raspberry Pi	